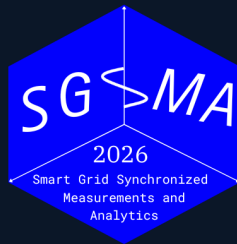


5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*



Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

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Universidad de los Andes | SGSMA 2026

Santiago de Chile · June 1–4, 2026

Date

June 1-4, 2026

Venue

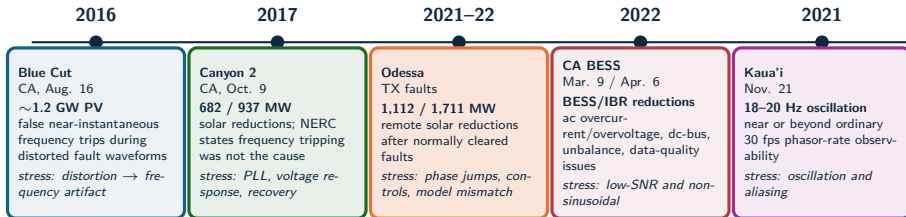
Santiago, Chile



Motivation

Why frequency estimation in low-inertia IBR grids is hard

Field events define the benchmark stress model



Benchmark implication: the estimator must be tested against waveform distortion, phase jumps, control recovery, unbalance, noise, harmonics, interharmonics, and reporting-rate limits, not only steady-state frequency error.

Source: NERC Blue Cut 2017; NERC/WECC Canyon 2 2018; NERC Odessa 2021/2022; NERC/WECC CA BESS 2023; Dong et al. Kaua'i analysis

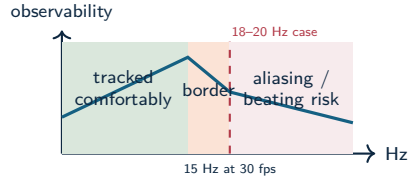
Standard PMU streams show consequences, not always causes

Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

Oscillation near reporting-rate limits

The Kaua'i Island event on Nov. 21, 2021 exhibited 18–20 Hz oscillations after an oil-power-plant trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.



Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.

Source: Dong et al., Kaua'i 18-20 Hz oscillation analysis, NERC IBR monitoring white paper, IEC/IEEE 60255-110-1

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

Operating claim

- Frequency estimation under grid disturbances is a **multi-objective measurement problem**.
- Rankings change with the objective: frequency RMSE, RoCoF error, latency, CPU cost, and disturbance sensitivity produce different orders.
- IBR-era disturbances make this operational: distorted waveforms and fast controls can turn a measurement artifact into a ride-through or forensic diagnosis problem.
- Estimator choice is conditional: event family, objective metric, and compute budget determine the preferred method.

Measurement rule

Report error, transient exposure, settling, latency, and cost as separate outputs for each estimator.

Technical contributions

Not claimed here

- PMU time-stamp certification;
- a single universal leaderboard;
- hardware latency measurements.

Contributions

- OpenFreqBench: a reproducible stress-test benchmark and platform for relay-class estimators;
- a metric-complete comparison protocol: RMSE, peak error, RoCoF error, T_{trip} , settling, and CPU;
- evidence that standard compliance tests miss IBR failure modes;
- an adapted RA-EKF as a tested estimator, with explicit $\dot{\omega}$ state and event gating.

The benchmark explains why estimator rankings change across objectives, regimes, and compute budgets.

Experiment scale

18
estimators

5
scenarios

4
families

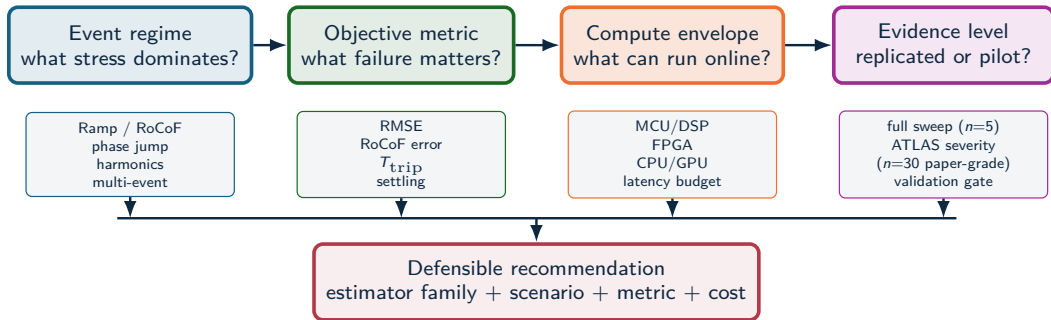
grid
search

Central finding

No single estimator wins across objectives: the ranking changes with scenario, metric, and compute budget. Protection-class frequency measurement needs tests beyond compliance-style cases.

Dataset: 18 estimators (incl. legacy baselines) across 33 scenarios, $n = 5$ Monte Carlo per scenario, with standard and IBR-specific metrics, plus ATLAS severity sweeps.

Estimator selection logic

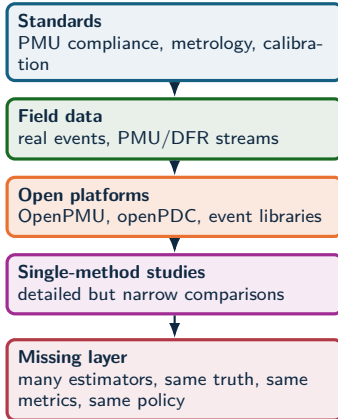


Example: RA-EKF is recommended for ramp/phase-jump protection metrics under DSP/FPGA-class cost; global RMSE leadership remains scenario-dependent.

Benchmark & platform

What is missing in prior work — and OpenFreqBench

Benchmark literature gap



Why existing evidence is not enough

Standards define requirements; field events define realism; open PMU ecosystems provide infrastructure. None alone answers which estimator family is defensible under the same disturbance, tuning policy, metric vector, and compute budget.



Estimator-level benchmark: one artifact contract for algorithms, scenarios, seeds, metrics, and plots

Result: controlled synthetic truth isolates estimator behavior under reproducible stress while field events define realism.

Benchmark comparison axes

Evidence source	Known truth	IBR composite stress	Trip-risk	CPU/latency	Reproducible artifacts
IEC/IEEE-style compliance	Yes	Limited	No	No	Device-dependent
Field PMU/DFR data	No	Yes	Possible	No	Dataset-dependent
Single-method studies	Often	Partial	Partial	Partial	Often limited
Estimator benchmark	Yes	Yes	Yes	Yes	Yes

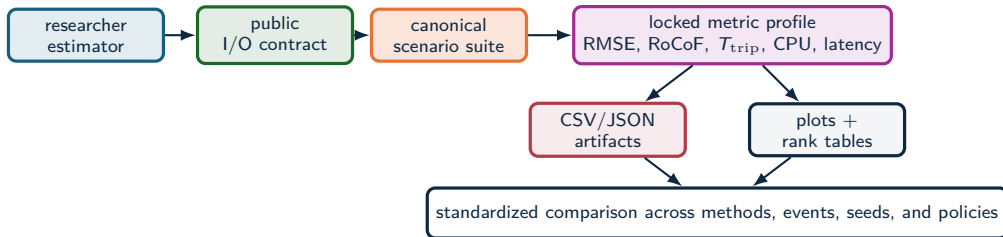
Measured outputs

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

Interpretive effect

An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.

OpenFreqBench platform



Researchers add

Estimator code, parameters, and hypotheses through the public contract.

Platform fixes

Truth signals, sampling, seeds, tuning policy, metric formulas, and schema.

Outputs

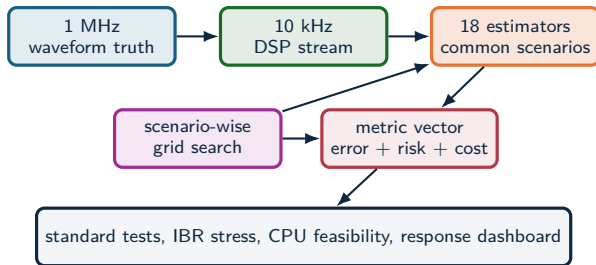
Benchmark reports, metric tables, plots, manifests, and readiness gates.

OpenFreqBench turns a new estimator into a repeatable benchmark entry instead of a one-off comparison.

Experimental method

Same truth, same metrics, fair tuning

Experimental method



Source: Methodology and test framework

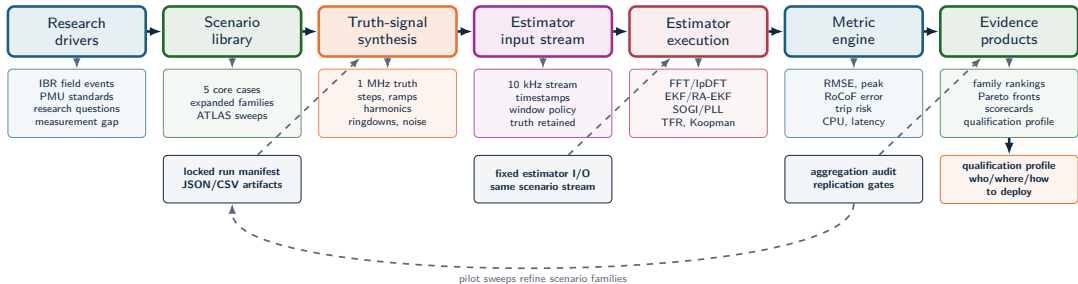
Measurement setting

Local, non-time-synchronized relay-class frequency tracking from a decimated 10 kHz stream; PMU timestamp certification is outside scope.

Experimental principle

Each estimator–scenario pair is tuned within a declared grid, then compared using RMSE, peak error, trip exposure, settling, and CPU.

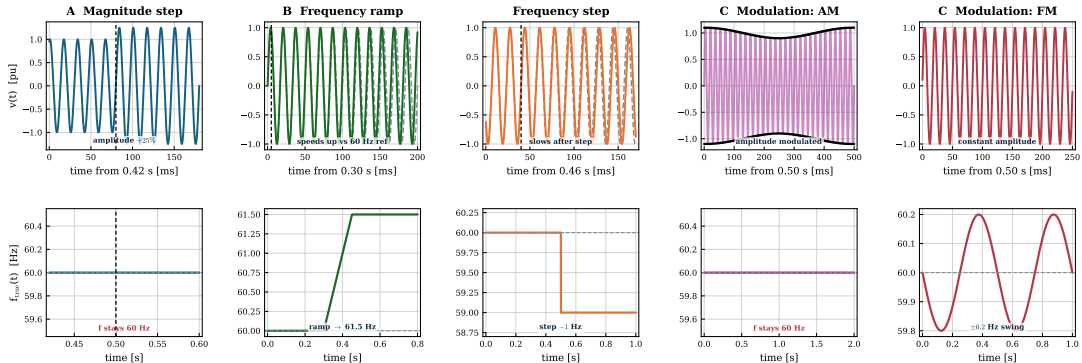
Full benchmark workflow



The benchmark is an evidence pipeline, not just a test script: every figure traces back to a locked manifest, a fixed I/O contract, and the readiness gate.

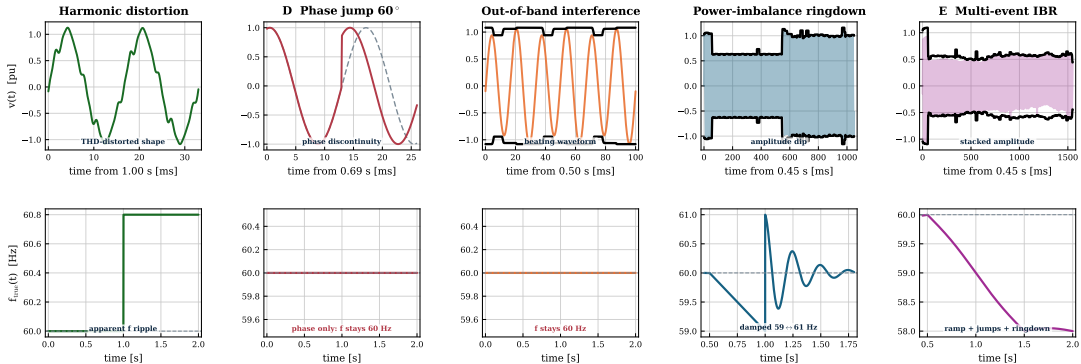
Scenario suite I — standard stresses

Scenario suite I — standard stresses: voltage $v(t)$ (top) vs. true frequency $f(t)$ (bottom)



Scenario suite II — composite IBR stresses

Scenario suite II — composite IBR stresses: voltage $v(t)$ (top) vs. true frequency $f(t)$ (bottom)



Estimator families

Loop-based

PLL, SOGI-PLL, SOGI-FLL,
Type-3 SOGI-PLL.

Model-based / stochastic

EKF, UKF, RA-EKF, LKF,
LKF2.

Legacy baselines

TKEO and ZCD are retained as
comparison references outside the
recommended deployment set.

Window / spectral

IpDFT, TFT, ESPRIT, Prony,
MUSIC.

Adaptive and data-driven

RLS, Koopman (RK-DPMU).

Interpretation rule

These families occupy different
latency and stress-sensitivity
positions. The comparison tracks
where each family wins, degrades,
and fails.

LKF and LKF2 follow H. Ahmed, S. Biricik, M. Benbouzid, "Linear Kalman Filter-Based Grid Synchronization Technique: An Alternative Implementation," IEEE, 2021.

Metrics and decision variables

Error and recovery

$$e[k] = \hat{f}[k] - f_{\text{true}}[k]$$

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_k e[k]^2}$$

$$E_{\text{max}} = \max_k |e[k]|$$

Protection-oriented risk

$$T_{\text{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5 \text{ Hz}\}$$

$$t_s = \min\{t : |e[\tau]| \leq \epsilon, \forall \tau \geq t\}$$

$$C_{\text{CPU}} = \text{mean time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

Core evidence

From standard cases to composite IBR stress

Standard scenario results

Method	Step	Ramp	Mod.
RA-EKF	0.000	0.006	0.037
SOGI-FLL	0.005	0.013	0.040
Koopman	0.003	0.017	0.044
TFT	0.001	0.019	0.046
UKF	0.000	0.017	0.055
LKF2	0.138	0.015	0.028
PLL	0.033	1.47	0.227
IpDFT	0.000	14.4	0.114
EKF	0.000	37.1	0.064

RMSE [Hz], mean over 5 seeds. Green:

below 0.05 Hz guide; red: failure/divergence regime.

What the heatmap shows

The standard step is easy for most methods. The ramp is the separator: EKF and IpDFT can look excellent at a point and still diverge in another seed.

Operational consequence

Robustness is not the same as point accuracy. A standard scenario can pass while the same estimator remains unsafe for ramp-like IBR dynamics.

Result: RA-EKF, SOGI-FLL, Koopman, and TFT remain below 0.05 Hz across the three standard cases; EKF and IpDFT fail through intermittent ramp divergence.

Scenario D: phase jump separates filters from windowed methods

State-space + loops recover cleanly

A clean $+60^\circ$ phase jump (NERC profile) is absorbed with near-zero error and **zero trip exposure**:

- EKF, UKF, RA-EKF: RMSE ≈ 0 Hz, $T_{\text{trip}} = 0$.
- PLL 0.021 Hz, SOGI-FLL 0.067 Hz, $T_{\text{trip}} = 0$.

Benchmark reading: an isolated phase jump does *not* break the state-space filters — EKF already recovers with zero trip exposure.

RA-EKF's real advantage appears on ramps and multi-event stress (next), not on clean phase jumps.

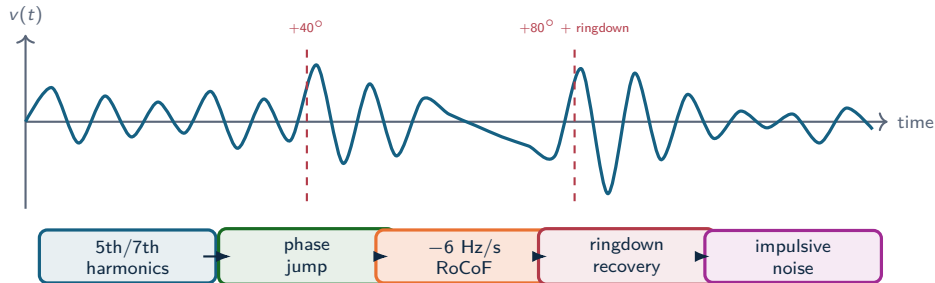
Source: full_mc_benchmark, $n = 5$: NERC_Phase_Jump_60

Windowed / spectral methods trip

The trip exposure here lives in the window/spectral family:

- IpDFT: 1.56 Hz RMSE, $T_{\text{trip}} = 1.21$ s.
- RLS 2.28 Hz / 1.30 s; Prony 2.39 Hz / 1.35 s.
- TFT 0.43 Hz / 0.09 s; Koopman 0.39 Hz / 0.11 s.

Composite IBR sequence: why it is hardest



Why standard tests miss it

The estimator sees spectral leakage, loop re-lock, state-model mismatch, transient recovery, and noise in one nonstationary waveform.

Why ATLAS decomposes it

The multi-event case proves the failure exists; severity sweeps identify which stressor actually breaks each estimator family.

Scenario E: the leaderboard breaks

Method	RMSE [Hz]	T_{trip} [s]	CPU [μ s]
UKF	1.09	0.78	14
EKF	1.13	0.87	11
SOGI-PLL	1.14	0.78	11
SOGI-FLL	1.14	0.84	9
IpDFT	1.14	1.35	17
RA-EKF	1.16	0.92	15
Koopman	1.90	0.84	15 241
ZCD	426	1.55	7

No accurate winner

Under the stacked sequence every usable method collapses to ≈ 1.1 Hz RMSE; ZCD diverges catastrophically (426 Hz). High accuracy is simply not available here.

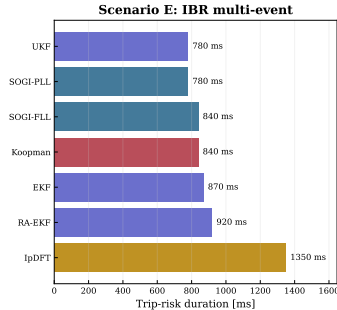
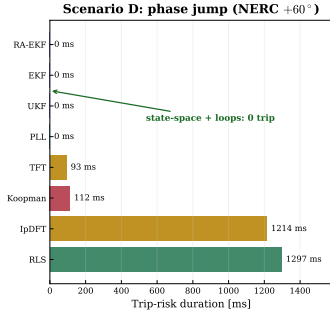
Source: full_mc_benchmark, $n = 5$: IBR_Multi_Event

Scalar ranking failure

UKF has the lowest RMSE and trip exposure, SOGI-FLL the lowest usable CPU, Koopman is $1000\times$ costlier; RA-EKF sits mid-pack but never diverges.

Trip-risk evidence: core stress cases

Trip-risk is an event metric, not an average-error metric



Result reading

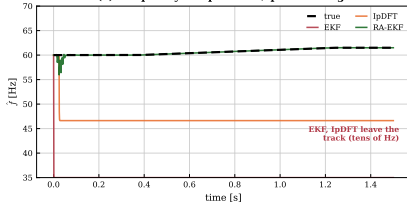
- Scenario D: state-space filters and PLL keep $T_{\text{trip}} = 0$; windowed/spectral methods (IpDFT, RLS) exceed 1.2 s.
- Scenario E: every method carries 0.8–1.4 s exposure — composite stress is unavoidable.
- The trip-exposure ranking is not the RMSE ranking.

Result: T_{trip} separates protection exposure from average error.

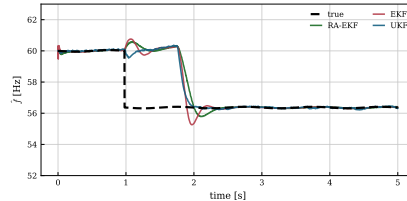
Event responses

Event responses (full_mc_benchmark): intermittent divergence is the dominant failure mode

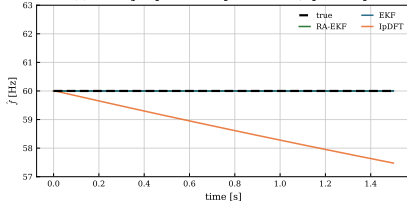
(a) Frequency ramp --- EKF/IpDFT diverge



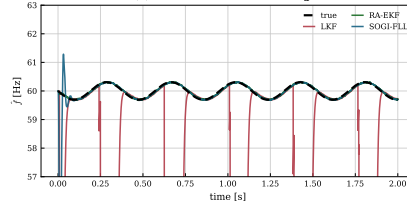
(b) Multi-event --- all degrade to ≈ 1 Hz



(c) Phase jump --- state-space recover, IpDFT trips



(d) Modulation --- LKF diverges



Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	State-space + SOGI-FLL near-zero; LKF2 best on mod	Step ≈ 0 ; mod 0.028–0.044 Hz	Clean cases do not predict ramp/event survival
Ramp and RoCoF tracking	RA-EKF is the most robust ramp tracker	0.006 Hz vs EKF 37 Hz, IpDFT 14 Hz (diverge)	Divergence is intermittent (seed-dependent)
Phase jump (Scenario D)	State-space filters recover with zero trip	EKF/UKF/RA-EKF $T_{\text{trip}} = 0$; IpDFT 1.21 s	Windowed/spectral carry the exposure
Multi-event RMSE	No accurate method — all ≈ 1.1 Hz	UKF 1.09 Hz best; ZCD 426 Hz diverges	RMSE hides trip-risk and 1000 $\times$ CPU spread
Embedded feasibility	PLL, SOGI-FLL, EKF, RA-EKF remain plausible	9–17 $\mu\text{s}/\text{sample}$; Koopman $\sim 15,000$	Python timing is a software-cost proxy

Selection changes with event family, objective metric, and compute envelope. A single leaderboard hides the deployment trade-off.

Computational feasibility

Method	Time/sample	Family	Target
ZCD	7 μ s	Loop	MCU
SOGI-FLL	9 μ s	Loop	MCU
PLL	9 μ s	Loop	MCU
EKF	11 μ s	Model	MCU/DSP
UKF	13 μ s	Model	DSP
RA-EKF	15 μ s	Model	DSP
IpDFT	17 μ s	Window	DSP/FPGA
ESPRIT	$\approx$ 1,400 μ s	Window	CPU
Koopman	$\approx$ 15,000 μ s	Data-driven	CPU/GPU

Source: full_mc_benchmark, $n = 5$: mean m13_cpu_time_us

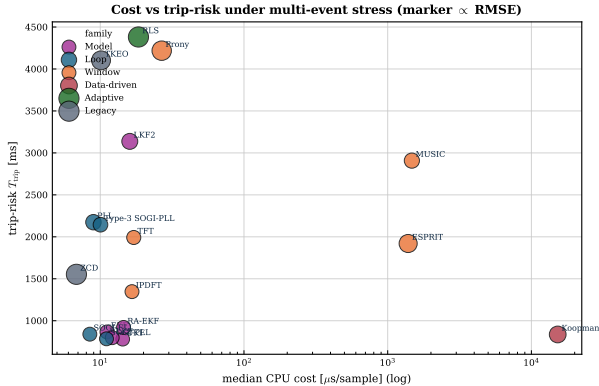
Deployment interpretation

The Python baseline separates feasible estimator families from clearly prohibitive ones; hardware latency requires platform timing.

Key comparison

The state-space and loop families all run at $\sim$ 10–17 μ s/sample (embedded-feasible); spectral (ESPRIT, MUSIC) and data-driven (Koopman) methods are 100–1000 $\times$ costlier and data-dependent.

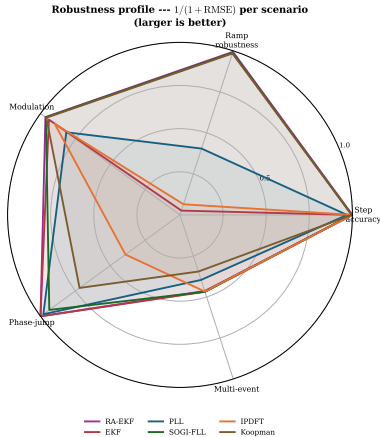
Deployment limits



Deployment interpretation

- Deployment combines RMSE, trip exposure, CPU, and divergence robustness.
- State-space + loop methods cluster at low cost ($\leq 17 \mu$ s) and low trip.
- Spectral/data-driven methods trade lower error for 100–1000 $\times$ higher CPU.
- RA-EKF stays low-cost with no ramp divergence in this run.

Balance profile



Radar profile

- Each axis is one scenario; larger = more robust ($1/(1 + \text{RMSE})$).
- RA-EKF and SOGI-FLL stay robust across all five scenarios.
- EKF and IpDFT collapse on the ramp axis (intermittent divergence).
- No estimator is strong on every axis — robustness is the discriminator.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Jump	Multi
Koopman	P	P	P	F	F
TFT	P	P	P	F*	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	P	P	P	F
PLL	P	F	F*	P	F
EKF	P	F	P	P	F
UKF	P	P	P	P	F
RA-EKF	P	P	P	P	F
LKF2	F*	P	P	P	F

Columns: **Jump** = phase jump (D), **Multi** = multi-event (E). **P** pass | **F** fails a declared RMSE/ T_{trip} /settling threshold | **F*** borderline.

Result: Step, ramp, and modulation compliance does not predict phase-jump or multi-event readiness.

Empirical claims

Protection behavior

Divergence, not phase jumps, is the dominant failure mode.
On the ramp, EKF and IpDFT intermittently diverge (37 and 14 Hz mean); RA-EKF stays at 0.006 Hz. A clean phase jump is recovered by every state-space filter with $T_{\text{trip}} = 0$.

Standard-test limitation

Simple-case pass/fail leaves event readiness unresolved.
The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

Guardrail: report scenario, metric, and compute envelope before naming a preferred estimator.

Metric disagreement

RMSE, trip-risk, and CPU select different winners.
Under multi-event stress UKF has the lowest RMSE/trip, SOGI-FLL the lowest usable CPU, and Koopman is 1000× costlier.

Deployability

Feasibility changes the accuracy ranking.
RA-EKF runs at $\sim 15 \mu\text{s/sample}$; the spectral and data-driven methods are 100–1000× costlier and sit outside relay-class timing.

Expanded & ATLAS

Does the pattern scale across families and severity?

Expanded stress analysis

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Monte-Carlo benchmark (full_mc)

- 18 estimators across 5 families.
- 33 scenarios (A–E plus severity variants).
- $n = 5$ Monte Carlo seeds per scenario.
- Standard, phase-jump, multi-event, and CPU metrics.

ATLAS severity sweeps

- Severity, sign, and policy curves per stress family.
- Fixed-policy, paper-grade $n = 30$ across all 9 required sweeps (both signs).
- Harmonic, RoCoF, phase-jump, and noise atlases.
- Tests sensitivity, scaling, and family stability.

Two evidence layers on the same truth: a replicated Monte-Carlo benchmark, plus ATLAS severity sweeps that trace where each estimator family degrades.

Expanded benchmark: family winners

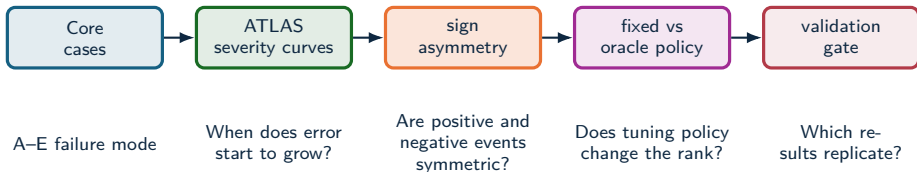
FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Family	Count	RMSE leader	RMSE	Trip-risk leader
IEEE ramps	9	RA-EKF	11 mHz	0 s (state-space)
Magnitude steps	7	UKF	<1 mHz	0 s (state-space)
Modulation	3	LKF2	15 mHz	0 s (state-space)
Phase jumps	3	UKF	<1 mHz	0 s (state-space)
Frequency step	1	RA-EKF	52 mHz	RA-EKF, 3 ms
OOB interference	1	UKF	<1 mHz	0 s (state-space)
IBR harmonics	3	SOGI-FLL	61 mHz	SOGI-PLL, 10 ms
IBR ringdown	4	LKF2	202 mHz	ESPRIT, 87 ms
IBR multi-event	1	UKF	1.09 Hz	UKF, 780 ms

The full Monte-Carlo sweep confirms the central pattern: the RMSE leader rotates (RA-EKF, UKF, LKF2, SOGI-FLL) by disturbance family, and error grows from sub-mHz on clean cases to ~ 1 Hz under multi-event stress.

ATLAS severity analysis

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Source: docs/ATLAS_PIPELINE.md; docs/ATLAS_METHOD_AUDIT.md

ATLAS result: metric choice changes the answer

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Comparison	Different winner	Levels tested
RMSE vs trip-risk	83	113
RMSE vs RoCoF max error	55	113
RMSE vs RoCoF RMS error	50	113

Winner concentration

RMSE is led mostly by EKF and ESPRIT. Trip-risk is led mostly by Koopman, EKF, SOGI-FLL, ZCD, and ESPRIT. CPU is always led by ZCD.

Result: across 113 severity levels, the preferred estimator changes when the objective changes.

Source: atlas-papergrade-missing-v1/global_metrics_report.csv

Why this matters

For protection-adjacent use, a low average error is not enough. The estimator must avoid long deadband exposure, track RoCoF, and remain computable.

Operational implication

The benchmark output is not a leaderboard. It is a metric-conditioned selection map.

ATLAS result: where estimators stop being valid

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Stress family	Maximum tested level	Methods below 0.05 Hz	Interpretation
Magnitude step	$\pm 90\%$ sag/swell	2	EKF and RA-EKF survive both signs.
AM modulation	10 Hz envelope	10	AM alone is not the hardest modulation case.
FM modulation	10 Hz modulation	0	FM is qualitatively harder than AM.
RoCoF ramp	± 50 Hz/s	0	Severe RoCoF requires a separate qualification class.
Phase jump	$\pm 60^\circ$	0	Single jump severity already breaks the 0.05 Hz guide.
Harmonics	30% THD	0	Low-THD winners do not survive extreme THD.
Interharmonics	20% at 75 Hz	0	Integer-harmonic tests are not sufficient.
Broadband noise	$\sigma = 0.1$ pu	0	Noise robustness is a first-class stress dimension.

AM, FM, and interharmonics split the story

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Stress	Winner pattern	Threshold behavior
AM modulation	EKF wins slow, reference, fast, and severe regions.	Ten methods stay below 0.05 Hz through 10 Hz envelope modulation.
FM modulation	TFT leads slow; ESPRIT leads reference/fast/severe; SOGI-PLL reaches the best threshold.	No method stays below 0.05 Hz at 10 Hz FM modulation.
Interharmonics	EKF wins trace, reference, high, and severe levels.	EKF crosses the 0.05 Hz guide around 8%; most other families cross earlier.

Implication

AM, FM, integer harmonics, and interharmonics should not be collapsed into one “distortion” label. They select different estimator families.

Decomposition insight

A composite IBR trace can contain AM, FM, harmonics, interharmonics, noise, and phase steps. ATLAS isolates which stressor changes the ranking.

When the textbook filter diverges

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario

Per-seed frequency RMSE [Hz] on the IEEE frequency-ramp scenario (5 Monte Carlo seeds):

Estimator	seed 1	seed 2	seed 3	seed 4	seed 5	trip-risk
EKF (textbook)	0.016	63.7	0.017	121.5	0.027	up to 1.35 s
RA-EKF	0.0055	0.0062	0.0083	0.0045	0.0058	0 s

What happens

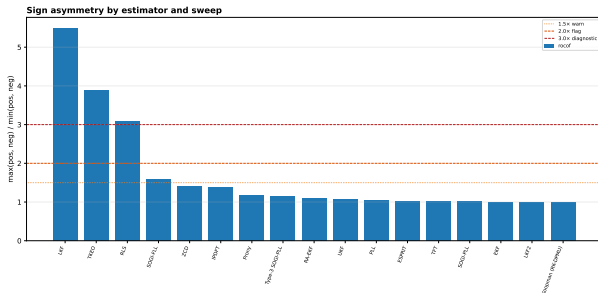
The standard EKF tracks most seeds but **intermittently diverges** to tens of Hz. It is *not* a severity threshold: the worst seed (121 Hz) had a low ramp rate (1.7 Hz/s) while the steepest (5 Hz/s) tracked cleanly. The failure is triggered by onset/phase combinations — seed-dependent and silent.

Why RA-EKF matters here

Covariance scaling and event gating remove the divergence: stable on **every** seed, zero trip exposure. This — not a universal RMSE win — is its defensible value.

RoCoF has a sign: down-ramps are harder

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



Source: atlas-papergrade-missing-v1/atlas_sign_asymmetry.csv ($n = 30$, both signs)

Directional bias ($RMSE_- / RMSE_+$ at ± 50 Hz/s)

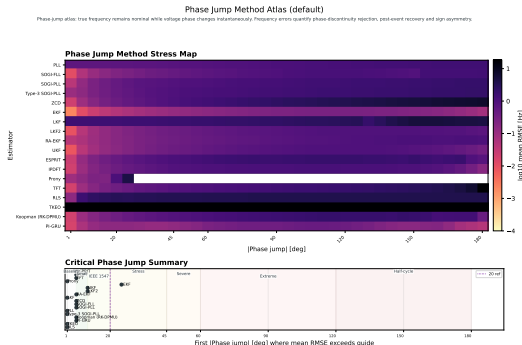
- LKF **5.5×** worse on $-RoCoF$
- TKEO **3.9×**, RLS **3.1×**
- SOGI-FLL **1.6×**
- EKF, UKF, RA-EKF ≈ 1.0 (symmetric)

Why it matters

A $+RoCoF$ -only test can certify a method that fails on frequency *decline* — the protection-critical direction. Both signs must be swept.

Phase-jump severity is non-monotone

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Worst case is not 180°

Most methods peak near 90–120° and *recover* toward 180° (a half-cycle jump partly self-cancels). EKF peaks at only **0.25 Hz @ 115°**, back to 0.07 Hz @ 180°.

The TFT cliff

One exception: TFT rises gently, then jumps $2.5 \rightarrow 21.9$ Hz between 150° and 180° — a phase-reversal singularity.

Result: Severity is non-monotone; sweep the jump angle rather than qualify one nominal point.

No universal winner: champions rotate by regime

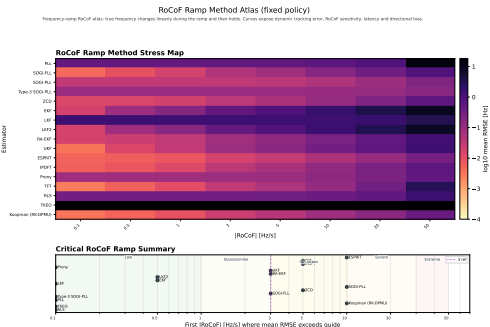
PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

RoCoF regime	Best	med. RMSE
Low (0.1–1 Hz/s)	Koopman	4.1 mHz
Standard (1–3)	ESPRIT	7.6 mHz
IBR stress (3–10)	ESPRIT	22.3 mHz
Severe (10–30)	SOGI-PLL	70.4 mHz

Across stress types

EKF wins phase jumps, the magnitude step and interharmonics; ESPRIT/Koopman win the RoCoF and frequency-step regimes; ZCD wins low-THD harmonics; SOGI-PLL leads severe RoCoF. No method leads everywhere.

Result: The deliverable is a selection map: event class $\times$ metric $\times$ compute budget.



Voltage sags and swells are not symmetric

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Estimator at $\pm 90\%$	Sag to 0.1 pu	Swell to 1.9 pu
EKF RMSE	47 mHz	14 mHz
RA-EKF RMSE	49 mHz	30 mHz
ZCD RMSE	374 Hz	9.4 mHz
ZCD peak error	2856 Hz	32 mHz

Mechanism

Zero-cross timing error scales roughly as $\sigma_n / (A\omega)$. A deep sag lowers zero-cross slope and makes false crossings likely when absolute noise is fixed.

Source: atlas-papergrade-missing-v1/global_metrics_report.csv; src/estimators/zcd.py

What to claim

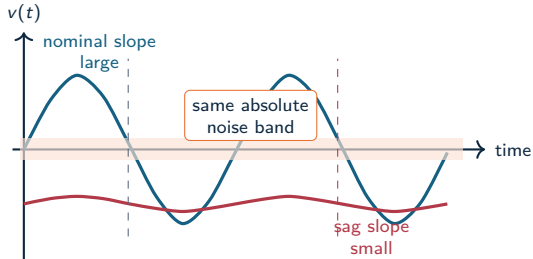
ZCD is amplitude-invariant only in the noise-free ideal case. Under fixed absolute noise, deep voltage sags create a low-SNR crossing problem.

Benchmark consequence

Sag and swell signs must be swept separately; a positive-only magnitude-step test can hide the worst case.

Why a voltage sag breaks zero-crossing detection

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)



Timing sensitivity

$$\Delta t \approx \frac{n(t_0)}{\left. \frac{dv}{dt} \right|_{t_0}} \quad \left. \frac{dv}{dt} \right|_0 \approx A\omega$$

When the voltage amplitude A collapses, the same noise produces much larger crossing-time error.

Estimator consequence

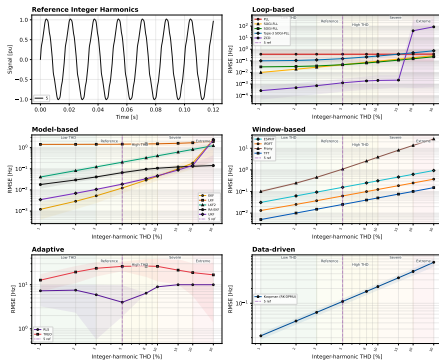
ZCD is elegant at clean high-SNR crossings, but deep sags convert amplitude loss into apparent frequency jitter and false crossings.

Harmonic degradation differs by estimator family

PAPER-GRADE SWEEP: $n = 30$, fixed policy, both signs — full-ATLAS readiness met (9/9 sweeps)

Integer Harmonics: RMSE [Hz] by estimator family (fixed policy)

Integer harmonic lab sweep: the true frequency is fixed at nominal and integer harmonic coefficients are added to trigger THD. Interharmonics, subharmonics, impulses and frequency events are disabled so degradation can be attributed to harmonic distortion rather than mixed, ill-defined effects.



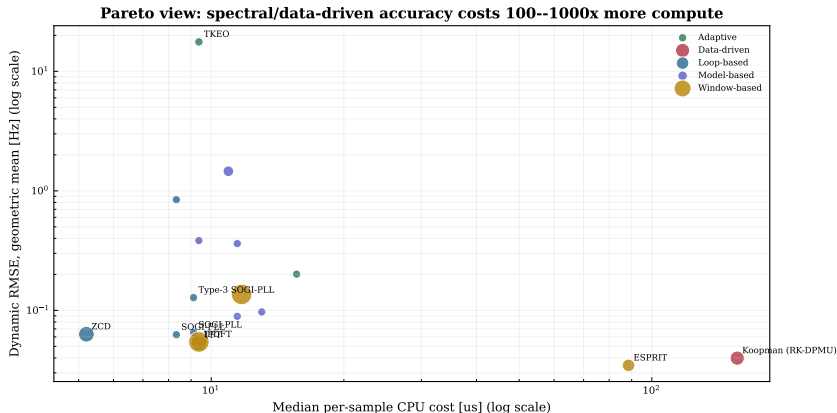
Shaded labels mark perturbation signs where applicable. Bands show available run intervals, thresholds and regions are interpretation guides.

Result

- Loop, window, model, and adaptive families degrade with THD at different rates.
- Best low-THD accuracy does not predict high-THD survival.
- Loop methods (ZCD) win at low THD but collapse above ~15%; window methods (TFT) degrade gently and lead at extreme THD.

CPU-accuracy Pareto analysis

FULL MONTE-CARLO BENCHMARK: 18 estimators, 33 scenarios, $n = 5$ per scenario



Findings & deployment

What to deploy — and what comes next

Three findings

1. **Estimator choice is metric-conditioned:** across ATLAS severity levels, the RMSE leader often differs from the trip-risk and RoCoF leader.
2. **Stress boundaries matter:** AM modulation remains survivable for many methods, while severe FM, RoCoF, interharmonics, noise, and extreme THD leave no method below the 0.05 Hz guide.
3. **Robustness beats point accuracy:** EKF, UKF and IpDFT intermittently diverge on ramps and multi-event stress, while RA-EKF's explicit $\dot{\omega}$ state keeps the ramp family stable across seeds.

Estimator qualification includes event family, sign, severity, transient recovery, trip exposure, compute feasibility, and standard-case RMSE.

Validity limits and next steps

Validation limits

- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

Validation program

- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with validation gates.

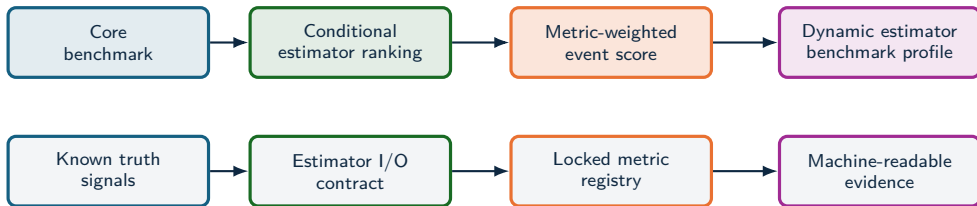
These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.

Recommendation map



Deployability depends on event class, trip-risk tolerance, and compute budget.

Qualification profile



Estimator selection moves toward **qualification** when each method is tested against a declared event profile, metric profile, and latency-cost envelope.

Standardization profile

Current standardization layer

- PMU standards define phasor, frequency, and RoCoF measurement requirements.
- Compliance tests are necessary, with estimator-family ranking by IBR event survival left outside scope.
- Device reports often hide algorithmic failure modes behind aggregate accuracy.

Standardization boundary

A dynamic-estimator profile needs declared signals, estimator I/O, metrics, evidence artifacts, and latency-cost reporting.

Benchmark profile

- Canonical events: steps, ramps, phase jumps, ringdown, harmonics, interharmonics, noise, and composite IBR sequences.
- Locked metrics: RMSE, peak error, RoCoF error, trip-risk, settling, invalid samples, CPU, and latency.
- Task profiles: forensic monitoring, PMU augmentation, DFR support, and relay-adjacent screening.

Standardization value

A reference test profile needs the same signals, estimator I/O, metrics, and machine-readable evidence.

IBR Event Qualification Score

COMPOSITE EVENT SCORE: weighted event-class qualification metric

Score for estimator e

$$\text{IBR-ERS}(e) = 1 - \sum_{s \in \mathcal{S}} w_s \Phi_s(e),$$

$$\Phi_s(e) = \alpha \tilde{E}_s + \beta \tilde{P}_s + \gamma \tilde{T}_s + \delta \tilde{S}_s + \eta \tilde{C}_s + \lambda \tilde{L}_s.$$

- $\tilde{E}_s$: normalized dynamic RMSE.
- $\tilde{P}_s$: normalized peak deviation.
- $\tilde{T}_s$: trip-risk duration.
- $\tilde{S}_s$: settling and recovery penalty.
- $\tilde{C}_s, \tilde{L}_s$: compute and latency penalties.

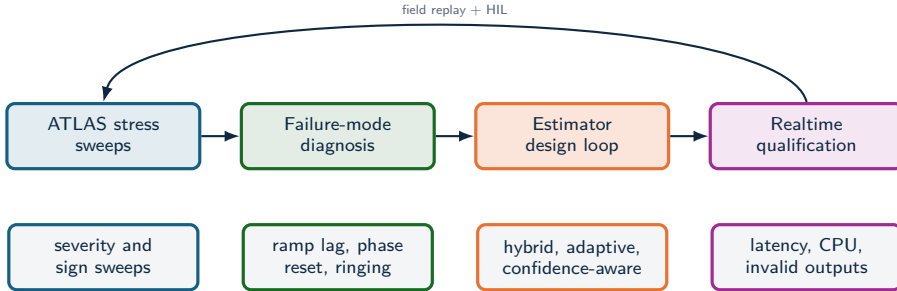
Composite metric

RMSE, trip-risk, transient recovery, and cost lead to different decisions. A composite score states the deployment objective explicitly.

Profile-specific weights

Protection-adjacent use can raise γ, δ, λ ;
forensic reconstruction can raise α, β ;
embedded monitoring can raise η .

Toward realtime IBR estimators



- The stress analysis identifies which physical stressor breaks each estimator family.
- That diagnosis supports hybrid estimators: event classifiers, harmonic-aware preprocessing, adaptive bandwidth, and self-reported confidence.
- Validation should combine synthetic truth, EMT replay, hardware-in-the-loop, and field oscillography.

Validation agenda and conclusion

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The study compares estimators across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Validation agenda

- Full ATLAS sweeps: fixed policy, 100-run Monte Carlo, statistical rank-stability analysis.
- Composite qualification profiles for forensic, PMU-augmentation, and relay-adjacent tasks.
- Three-phase, unbalance, and converter-control interaction coverage.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize the full objective vector: frequency error, RoCoF, trip exposure, settling, CPU, latency.

Result: estimator choice is tied to event class, metric target, and compute envelope; estimator qualification must declare the stress regime, latency limit, and evidence depth.